

# Kamran Heydarov

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## SUMMARY

Chemoinformatics M.Sc. student working at the interface of physical chemistry, molecular modelling and machine learning. Experienced across the full QSPR workflow: curating experimental data from chemical databases, computing and selecting molecular descriptors, training and validating ensemble models, and deploying them as documented tools that non-specialists can use. Author and maintainer of open-source scientific Python libraries for ionic liquid thermodynamics and thermophysical data access (pyionics, ILThermoGUI, ILQSPR), working in Python with RDKit, Mordred, scikit-learn and XGBoost, complemented by quantum chemistry coursework and DFT calculations. Particularly interested in applying data-driven methods to the rational design of ionic liquids and sustainable solvents.

## EDUCATION

### University of Strasbourg

M.Sc. in Chemoinformatics

Baku, Azerbaijan

Expected Jun 2027

### Azerbaijan State Oil and Industry University

B.Sc. in Chemical Engineering, Chemical Technologies Faculty

Baku, Azerbaijan

Jun 2025

- Graduation thesis, Department of Chemistry and Inorganic Substances Technology ([ResearchGate](#)).

## WORK EXPERIENCE

### Azerbaijan State Oil and Industry University

Django Developer & Chief Specialist

Baku, Azerbaijan

Sep 2025 – Jun 2026

- Engineered Django applications and managed database schemas (e.g., MongoDB) to handle critical university data.
- Built and maintain the university's scholarly publishing platforms, front end and back end: [press.asoiu.edu.az](https://press.asoiu.edu.az) and the OJS journal system at [ojs.asoiu.edu.az](https://ojs.asoiu.edu.az).
- Coordinated the migration of *Baku Mathematical Journal* to Elsevier Digital Commons across nine international working meetings with the Elsevier team, covering site structure, metadata preparation and article migration; serve as Executive Editor of the journal.
- Registered DOIs and deposited metadata through Crossref, aligning the university's journals with international indexing requirements.
- Designed and programmed the official website of the MacroFrontiers 2026 international conference ([asoiu.edu.az/mpmc2026](https://asoiu.edu.az/mpmc2026)), and built the electronic submission, tracking and plagiarism-screening system for the NASCO international conference.
- Administered Elsevier research platforms (Scopus, ScienceDirect, SciVal, Reaxys) and Turnitin for ASOIU and UFAZ, and delivered SciVal and Reaxys training seminars to academic staff.

### Azerbaijan State Oil and Industry University

Data-Oriented Lab Assistant, supervised by Dr. Leyla Vazirova

Baku, Azerbaijan

Sep 2023 – Feb 2025

- Assisted professors in undergraduate laboratories at UFAZ and ASOIU.
- Guided 20+ students per lab session, improving their understanding of experimental concepts and helping boost lab grades.
- Taught students statistical methods for chemical analysis, enhancing their data interpretation skills and analytical reasoning.

## PROJECT EXPERIENCE

### ILQSPR: QSPR Modelling of IL–Solvent Mixtures ([code](#), [app](#), [docs](#))

Baku, Azerbaijan

Data Scientist, joint work with Dr. Shamkhal Baybekov ([email](#))

2025 – Present

- Built QSPR models predicting the density and refractive index of pure ionic liquids and their binary mixtures with water, ethanol, and isopropanol as a function of temperature and IL mole fraction.
- Curated training data from the NIST ILThermo database using pyionics, applying duplicate removal, consistency checks, and structure standardisation.
- Represented ionic liquids with Mordred 2D descriptors computed separately for cation and anion, followed by variance- and correlation-based feature selection.
- Trained consensus ensembles of five XGBoost models using group-based 5-fold cross-validation on IL SMILES, reaching  $R^2 = 0.90\text{--}0.96$  across all eight property–solvent systems; manuscript in preparation.
- Deployed the models as a public Streamlit application with online documentation, so that non-specialists can obtain predictions directly from SMILES, temperature and composition.

### pyionics Python Library ([link](#))

Baku, Azerbaijan

Owner & Developer

Mar 2025 – Present

- Developed a Python-based tool for thermodynamic property estimation of ionic liquids and mixtures.
- Integrated models for phase equilibrium and heat capacity prediction to support research and education.
- Optimised the codebase for usability in academic and industrial thermodynamic analysis.

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| <b>marimo4obs, Obsidian Plugin (code)</b>                                                                                                                                                                                                                                                                                                                                                                                               | Baku, Azerbaijan    |
| <i>Owner &amp; Developer</i>                                                                                                                                                                                                                                                                                                                                                                                                            | 2026                |
| <ul style="list-style-type: none"> <li>Built an open-source Obsidian plugin that registers the .py extension and opens marimo reactive Python notebooks inside the vault, each backed by a local marimo server.</li> <li>Supports both the full reactive editor and app mode, and installs marimo automatically when missing.</li> <li>Distributed as a community plugin with packaged releases and documented installation.</li> </ul> |                     |
| <b>Chron Shift, Browser Game (play)</b>                                                                                                                                                                                                                                                                                                                                                                                                 | Baku, Azerbaijan    |
| <i>Creator &amp; Developer</i>                                                                                                                                                                                                                                                                                                                                                                                                          | 2026                |
| <ul style="list-style-type: none"> <li>Developed a single- and multiplayer browser platform game with running and fighting mechanics; selected with it as one of the top 15 studios for the Xsolla–IDDA Baku game incubator.</li> </ul>                                                                                                                                                                                                 |                     |
| <b>ChemGraph, Process Simulation Software</b>                                                                                                                                                                                                                                                                                                                                                                                           | Baku, Azerbaijan    |
| <i>Executive Director</i>                                                                                                                                                                                                                                                                                                                                                                                                               | Feb 2025 – 2026     |
| <ul style="list-style-type: none"> <li>Led the development of ChemGraph, an open-source PFD design simulator supported by Azerbaijan State Oil and Industry University.</li> <li>Oversaw its use in education and research, and extended its functionality to improve accessibility for academic and industrial applications.</li> </ul>                                                                                                |                     |
| <b>ILThermoGUI</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | Baku, Azerbaijan    |
| <i>Data Scientist, group of Dr. Shamkhal Baybekov (email)</i>                                                                                                                                                                                                                                                                                                                                                                           | Sep 2024 – 2026     |
| <ul style="list-style-type: none"> <li>Developed a Python library for installing and accessing NIST ILThermo data.</li> <li>Cleaned and structured the datasets, and designed visualisations and statistical analyses to interpret thermophysical data.</li> <li>Researched ionic liquids, focusing on their thermophysical properties and applications.</li> </ul>                                                                     |                     |
| <b>MentorCorp</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | Baku, Azerbaijan    |
| <i>Co-Founder &amp; CTO</i>                                                                                                                                                                                                                                                                                                                                                                                                             | Jan 2024 – 2026     |
| <ul style="list-style-type: none"> <li>Developed a user-friendly interface for the mentoring platform, streamlining resource access and reducing average onboarding time by 30% for new users.</li> </ul>                                                                                                                                                                                                                               |                     |
| <b>YOUNGH</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | Baku, Azerbaijan    |
| <i>Founder &amp; CEO</i>                                                                                                                                                                                                                                                                                                                                                                                                                | Jan 2024 – Sep 2024 |
| <ul style="list-style-type: none"> <li>Built a Q&amp;A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first six months.</li> <li>Enabled event tracking through the website, and sold the platform to a company in our competition.</li> </ul>                                                                                                                                              |                     |

## CONFERENCES & PRESENTATIONS

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Poster</b> S. Baybekov, K. Heydarov, O. Abdullayev, Kh. Gaybaliyeva, <i>QSPR Models for Predicting Density and Refractive Index of Ionic Liquid Binary Mixtures</i> (abstract). |                |
| Strasbourg Summer School in Chemoinformatics (CS3-2026), University of Strasbourg, France                                                                                          | 22–26 Jun 2026 |

## LEADERSHIP & OUTREACH

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |
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| <b>Science Club</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Baku, Azerbaijan |
| <i>Founder</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2023 – 2026      |
| <ul style="list-style-type: none"> <li>Founded and ran a student science club organising academic competitions and scientific outreach events.</li> <li>Organised Integration Bee 2023 Azerbaijan, an integral-calculus competition for university students.</li> <li>Created AnimAI, a prompt-engineering and generative-image competition, and co-organised GenerO with the UFAZ Chemical Engineers Club, a challenge on the synthesis of oxygen in the Martian atmosphere.</li> </ul> |                  |

## INCUBATION PROGRAMS

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|-------------------------------------------------------------|---------------------|
| <b>SABAH.HUB Incubation</b> Project Owner, YOUNGH           | Jan 2024 – May 2024 |
| <b>UFAZ IGNITE Incubation Program</b> Project Owner, YOUNGH | Oct 2023 – Jul 2024 |

## SKILLS & LANGUAGES

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|----------------------------------------------------------------------------------------------------------------------------|
| <b>Cheminformatics &amp; ML</b> RDKit, Mordred descriptors, scikit-learn, XGBoost, QSPR modelling, KNIME                   |
| <b>Computational chemistry</b> Density Functional Theory for molecular quantum dynamics, thermodynamic property estimation |
| <b>Programming</b> Python (pandas, NumPy, matplotlib, tkinter), Django, Streamlit, JavaScript, MongoDB, Git, $\LaTeX$      |
| <b>Other tools</b> MATLAB, SolidWorks, Excel                                                                               |
| <b>Languages</b> Azerbaijani (native), Turkish (native), English (fluent), Russian (B1), French (A2)                       |

## CERTIFICATIONS

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data &amp; programming</b> Data Manipulation with pandas, MongoDB Python Developer Path, Intermediate Python, JavaScript (Intermediate), GitHub Foundations, KNIME Fundamentals, Excel Fundamentals |
| <b>Process &amp; laboratory safety</b> SChE <sup>®</sup> Managing Lab Process Safety 1 & 2 (ELA982, ELA986), SChE <sup>®</sup> Certificate Program: Lab Safety (ELA954)                                |